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Rafeeq (رفيق): Development and Evaluation of a Wearable-Free, Smartphone-Based AI Platform for Seizure Forecasting, Detection, and Family Support in Paediatric Epilepsy in the UAE (Ages 6–12)

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Introduction

Paediatric epilepsy is a common condition affecting millions of children globally, but the uncertain nature of seizures results in an additional burden for the child and the caregiver [1][2]. Seizures can occur at school, at home during sleep time or in public places and therefore requires caregivers to be always ready to help[1][3]. In the UAE, it has been found that 1 out of 200 children suffers from epilepsy [4]; however, a systematic review conducted in 2023 has shown that there is a lack of local epidemiological data [5]. This uncertainty and continuous need for management can lead to delayed treatment responses, increased stress for caregivers, and heightened anxiety for both children and their families [6][3].

Current technologies are limited in scope when it comes to managing epilepsy. Wearable devices pose cost, discomfort, and adherence issues [7][8], while diary-based methods depend heavily on consistent input from caregivers, who often abandon them over time [9]. Other systems focus mostly on detecting seizures, tracking symptoms, or clinical review [10][11]. This highlights the need for a comprehensive approach that supports all three key stakeholders in paediatric epilepsy care the child, caregiver, and clinician within a single system.

The UAE is well positioned to pilot such an approach, with national digital-health infrastructure such as the Ministry of Health and Prevention's platform and the growing use of AI-supported clinical technologies [12]. To address these gaps, this study proposes "Rafeeq," a bilingual (Arabic–English), AI-powered paediatric epilepsy platform designed to provide continuous, wearable-free support using a child's smartphone. During the day, acoustic sensing estimates physiological signals such as heart rate, breathing rate, and heart rate variability (HRV) to support seizure-risk forecasting and seizure-related monitoring. During sleep, camera-based monitoring is complemented by the same physiological signals to support multimodal seizure detection. A personalized story companion, caregiver training module, and clinical dashboard connect the child, caregiver, and clinician within a unified care platform.

Research Questions

In this research study, Rafeeq, a device-free, AI-driven paediatric epilepsy platform designed to predict, detect seizures, and bring together the child, caregiver, and physician in one place, is to be developed. The following questions will be answered in this research study:

1. How effective is the Rafeeq app in terms of:
    a. Accuracy of sonar-based physiological sensing
    b. Accuracy and lead time of seizure risk forecasting
    c. Accuracy of day (sonar) and night (camera and sonar) seizure detection
    d. Speed of alert delivery to caregivers
2. How effective are Rafeeq's supporting modules in terms of Child engagement with the story companion.
3. Is there a significant difference in Rafeeq's core performance in terms of:
    a. Forecasting accuracy across seizure types
    b. Detection accuracy between day and night modes
    c. Alert latency across network conditions

4. Is there a significant difference in caregiver and clinician outcomes in terms of:
    a. Caregiver confidence before vs. after training
    b. Clinician review efficiency using the dashboard vs. traditional review
5. What is the rate of acceptability of Rafeeq in terms of:
    a. Functionality
    b. Durability/reliability
    c. Accessibility

Literature Review

37 Paediatric epilepsy is a major global health burden, with an estimated 8 million children and adolescents living with epilepsy as of 2021, with incidence particularly high in early childhood [1]. Beyond the seizures themselves, their unpredictability places substantial stress on caregivers and families [2][6]; interviews with parents of children with treatment-resistant epilepsy describe persistent fear surrounding the possibility of another seizure [3]. Existing tools do not fully address these challenges: reviews of wearable-based systems identify cost, discomfort, and device visibility or stigma as important barriers to adoption [7][8][9], while studies comparing seizure diaries with video-EEG recordings have reported substantial underreporting, with estimates ranging from 40–60%, particularly for seizures occurring overnight [10][11].

Passive, phone-based sensing is technically feasible. Several studies demonstrate that a smartphone's built-in speaker and microphone can function as active sonar, measuring breathing and heart rate contactless with a sub-1 bpm error rate relative to ECG references [11][14][15]. HRV is a well-established preictal biomarker: wearable HRV-anomaly detection systems achieve up to 85.7% sensitivity, while HRV-SVM classifiers exceed 89% sensitivity in distinguishing preictal from interictal states [16][17]. However, because preictal signatures vary across individuals and seizure types, patient-specific calibration is required [17][18]. For nocturnal monitoring, CNN/optical flow video systems detect convulsive seizures with accuracy comparable to wearables, and contactless systems evaluated against video-EEG (ages 0–80) show strong sensitivity and specificity across various seizure types [19][20], supporting the implementation of sonar-confirmed video over video alone.

Regarding user engagement, studies on paediatric health apps and gamified tools show that storytelling and narrative rewards are the most effective strategies for sustaining engagement [21][22]. In terms of caregiver readiness, seizure first-aid literature reports that more than half of caregivers lack confidence prior to training, though structured programs significantly
26 improve their self-efficacy [23]. On the clinical side, AI-assisted EEG analysis is approaching readiness as a supervised decision-support tool reducing review time without replacing clinician judgment [24], thereby supporting Rafeeq's dashboard as an assistive rather than autonomous tool.

Locally, UAE epilepsy affects roughly 1 in 200 children, managed through centres such as Al Jalila Children's Hospital's Paediatric Neurosciences Center of Excellence [4][25], yet dedicated local epidemiological data remains thin even as national digital-health infrastructure is a strategic priority [12] a combination that motivates piloting Rafeeq in the UAE specifically. Taken together, the literature supports an integrated model current tools lack wearable-free

sensing, evidence-based engagement, and assistive clinical review, while flagging HRV variability, sensor robustness, and calibration needs as design priorities.

Proposed Methodology

The method is a quantitative approach.

1. **Platform Development:** Rafeeq is a bilingual (Arabic & English) mobile application involving only the smartphone of the kid without a need for any wearable device. The speaker and microphone incorporated within the smartphone serve as the active acoustic sonar that helps determine breathing rate, heart rate and HRV during the day while the front camera of the smartphone serves as the video monitor during the night, validated by the same sonar signal to avoid any false alarms. Rafeeq also includes story driven companion and caregiver training modules (videos and quizzes).
2. **AI-Driven Risk Forecasting and Seizure Detection**: The AI-driven pipeline that Rafeeq has designed contains two components, seizure-risk forecasting, and non-contact seizure detection. The seizure risk forecast pipeline involves the use of information related to the seizure history of the child, adherence to the medication plan, sleep quality of the child, identified triggers, and any coexisting conditions. Using machine-learning algorithms like the Random Forest and XGBoost, the risk forecasting model produces a risk forecast. Explainable AI helps identify the factors that are responsible for individual risks. The non-contact seizure detection algorithm involves estimating the physiological signals of heart rate, breathing rate, and heart rate variability (HRV). This is done using contactless acoustic sensing during the day and using camera-based monitoring and continuous acoustic sensing at night. The detected seizures trigger caregiver alerts. The design of this system ensures that the physiological and video data collected are stored in the device without needing constant cloud upload.
3. **AI-Powered Child and Caregiver Support:** In addition to clinical prediction through AI, Rafeeq uses it for child and caregiver assistance purposes. After a seizure has occurred, the Recovery Story Companion creates a personalized age-appropriate story using the child's input either in Arabic or English with the help of NLP and an LLM, which is read to the child using text-to-speech technology. The system has safeguards to review any disclosures that may arise from the child's responses and refer them to the caregiver or the appropriate human professional, rather than responding directly to them via the AI.
4. **Clinical Dashboard:** The clinical dashboard allows the authorized physician to have a role-based unified longitudinal view of the health history of the patient, including the history of seizures and risks, diary information of the child or caregiver, medication adherence, and other input. The neurologists can upload EEGs that will be reviewed with the help of AI, which highlights parts that may be clinically significant. The results of the AI are indicated as decision support guidelines only.

Testing and Evaluation

The Rafeeq system will be subject to both bench and field testing to verify that the platform is reliable and precise. At first, both types of testing will be carried out to ensure the validity of each core function separately prior to integrated field application.

Bench testing will involve checking the accuracy of heart rate, breathing rate, and HRV readings of the sonar-based sensing module in comparison with reference medical-grade

devices (such as a pulse oximeter and an ECG) at different distances, phone positions, and sound volumes. For the CNN-based detection models, the accuracy will be verified based on recorded/labelled audio and video datasets of seizures and non-seizures and measured by such metrics as sensitivity, specificity, and false alarm rate.

For field testing, it will be carried out using the system for the selected volunteer families from a partner paediatric neurology clinic, with emphasis on:

- Performance and accuracy of detection under normal home conditions (background noise, presence of other individuals in the environment, diverse room set-ups)
- Timing of alert, time between the detection of the seizure risk and informing the caregivers
- Reliability of alert under various connection situations (Wi-Fi, cellular data)
- Switching from day to night mode reliability, impact on the performance of the phone of the child

Acceptability Rating

The caregivers, paediatric neurologists, and sometimes the children themselves will assess the acceptability of Rafeeq on ISO-based 4-point criteria. The assessment will be on three aspects:

- **Functionality**: that the sensing, prediction, detection, alarm, storytelling, training, and dashboard functions work well as designed.
- **Reliability/durability:** the consistency of the system's performance, including how well it deals with missed detections, false alarms, and connectivity issues.
- **Accessibility/Ease of use:** the ease with which the system can be installed and set up, along with clear bilingual (Arabic and English) instruction.

Performance Evaluation and Recommendations

The performance evaluation of the entire Rafeeq system will be measured by detection accuracy, forecasting reliability, alert responsiveness, and user acceptance. The efficacy and effectiveness of Rafeeq will be statistically assessed using the Arithmetic Mean, ANOVA, Weighted Mean, and the Likert Scale. Calibration will be individually evaluated for detection thresholds, sonar and camera sensing performance, alert transmission, and mobile app usability. Recommendations will focus on enhancing Rafeeq's reliability for paediatric epilepsy monitoring across functionality, durability, and accessibility, with particular emphasis on patient-specific model calibration and system performance within the UAE's bilingual household context.

Expected Outcome

The study is expected to demonstrate the effectiveness of Rafeeq as a low-cost, wearable-free, AI-powered paediatric epilepsy monitoring platform. It will evaluate the accuracy of sonar-based physiological sensing and CNN-based detection, the reliability of day/night seizure forecasting and detection, and the usability of the story-driven companion, caregiver training module, and clinical dashboard. Results are expected to guide improvements in accessible paediatric epilepsy monitoring, offering earlier, explainable warning of seizure risk, faster caregiver response, higher daily engagement through story-driven design, and a continuous case history that speeds clinical review for neurologists.

Potential Limitation

Due to variability in ambient household noise and physical obstructions (e.g., blankets, distance from the phone) affecting sonar-based sensing accuracy, and due to variable network conditions affecting alert transmission speed, the system may experience occasional false positives, missed detections, or delayed alerts, particularly in noisy environments or areas with limited connectivity. Additionally, preictal physiological signatures are known to vary meaningfully between individuals and seizure types, meaning forecasting accuracy may require ongoing, patient-specific model calibration rather than a single generalized threshold.

Conclusion

Rafeeq offers a practical, wearable-free solution for paediatric epilepsy monitoring, forecasting, and family support. By integrating sonar-based physiological sensing, CNN-based day/night seizure detection, AI-driven risk forecasting, and a story-driven engagement layer with a clinician-facing dashboard, the system aims to replace panic with warning and isolation with connection, supporting children, caregivers, and neurologists in the UAE and beyond as one continuous, confidence-building system.

Project Practicalities

The project is expected to run for around 6 months. The first month will cover requirements gathering, architecture planning, preparation of datasets, and establishment of clinical partnerships and protocols for testing. The second and third months will cover development and training of the seizure risk prediction and detection models as well as the development of the mobile application which includes Recovery Story Companion and Caregiver Training Modules, as well as the development of the clinical dashboard. The fourth month will be for integration of AI models with mobile application and clinical dashboard, followed by bench testing of the system components individually. The fifth month will be devoted to conducting field tests of the integrated system under household conditions, collecting system performance and usability data, and refining the system based on the collected data. The sixth month will involve data analysis, system and evaluation results refinement, and reporting of the project findings.

Post-program Plan

Upon completion, the researchers intend to refine Rafeeq through additional field validation with a larger patient cohort and present the findings at academic and health-innovation conferences or publish them in a relevant journal. Partnerships with UAE paediatric neurology centres and potential integration with national digital-health infrastructure are being considered for future pilot implementation.

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